Sec 11.6.

Lecture 27: Cylinders and Quadric Surfaces

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

Sem II, 2024/2025

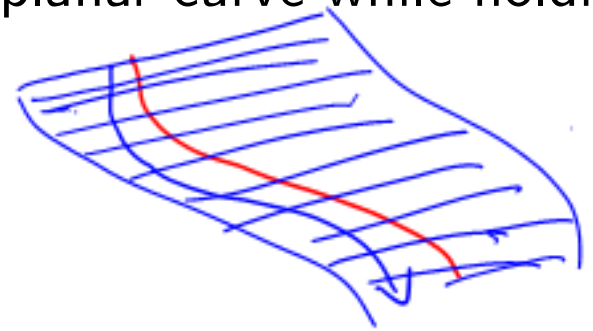
Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Identify cylinders based on their equation.
- ☐ Identify standard quadric surfaces based on their equations.

Cylinders

A cylinder is a surface generated by moving a straight line along a given planar curve while holding the line parallel to a given fixed line.



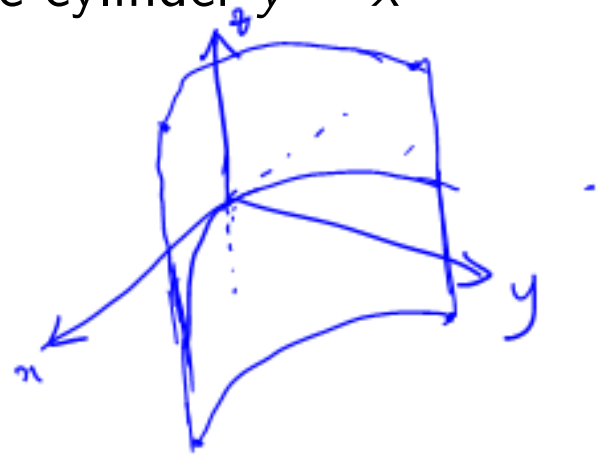


Can be represented by an equation missing one of the variables.

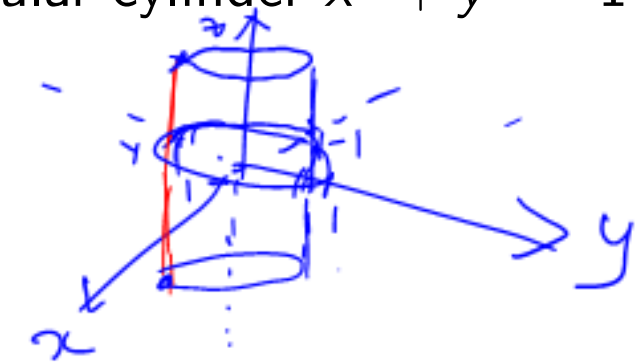
$f(x,y)$
$g(y,z)$
$h(x,z)$

Example:

Parabolic cylinder $y = x^2$



Circular cylinder $x^2 + y^2 = 1$



Level Curves

A level curve (or contour) of a function is a curve parallel to xy-plane corresponding to the equation of the form $f(x, y) = c$, where c is a constant.

For each level curve, the corresponding value of c gives the height of the curve above (or below) the xy-plane.

Example: ~~Sketch~~ Sketch the level curves for the function $f(x, y) = x^2 + y^2$.

Let $x^2 + y^2 = c$ $\quad x^2 + y^2 \geq 0 \Rightarrow c \geq 0$

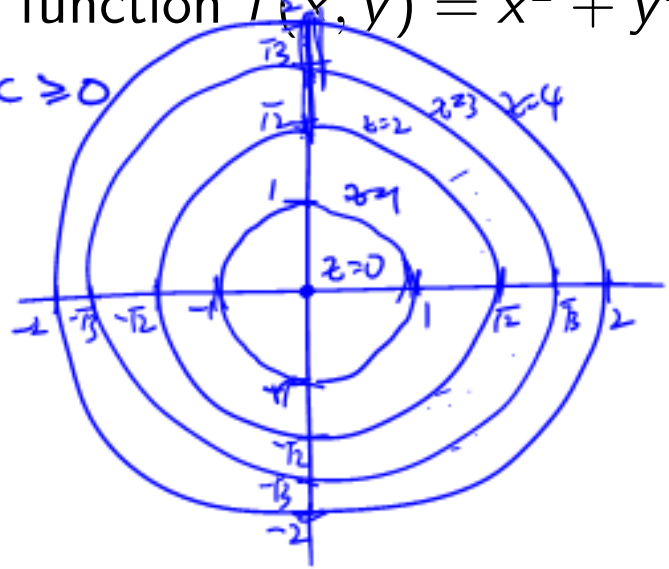
At $z = c = 0$, $x^2 + y^2 = 0$

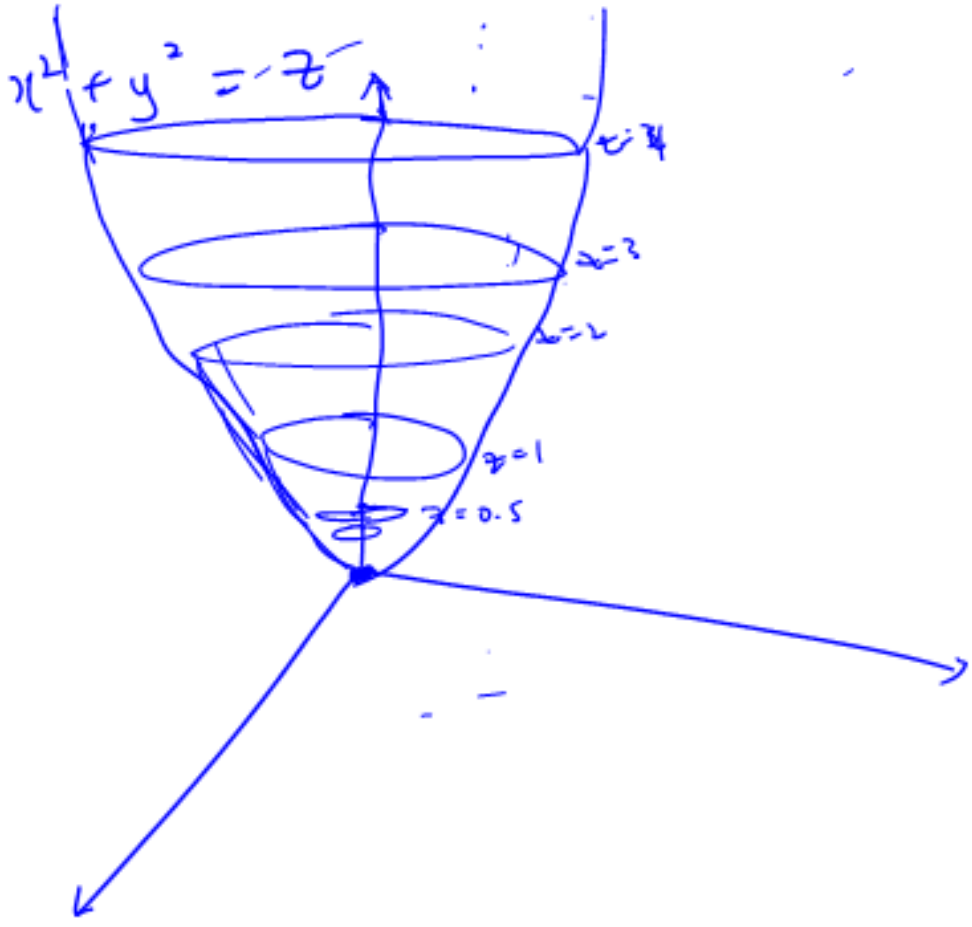
At $z = c = 1$, $x^2 + y^2 = 1$

At $z = c = 2$, $x^2 + y^2 = 2$

At $z = c = 3$, $x^2 + y^2 = 3$

At $z = c = 4$, $x^2 + y^2 = 4$




$x^2 + y^2 = z$
$z = 3$
$z = 2$
$z = 1$
$z = 0.5$

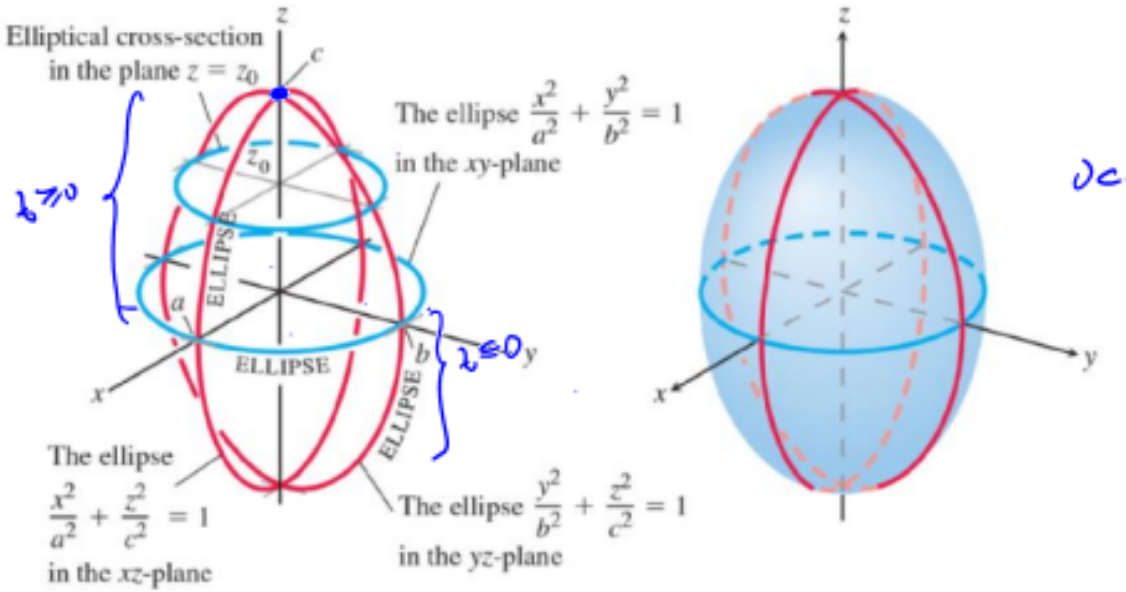
Quadric Surfaces

A quadric surface is the graph in space of a equation of the general form

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0$$

where $A, B, C, D, E, F, G, H, I, J$ are constants.

Ellipsoid

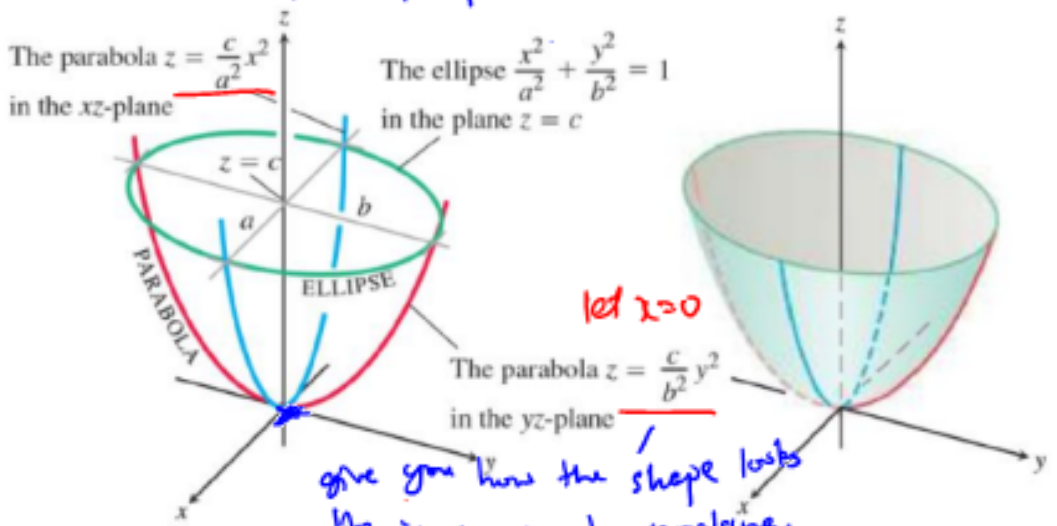


ELLIPSOID $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

Quadric Surfaces

Elliptical Paraboloid

let $y=0$ → give you how the shape looks like in parallel to xz-plane.



let $x=0$ → give you how the shape looks like in parallel to yz-plane.

ELLIPTICAL PARABOLOID $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} \geq 0$, $\frac{z}{c} \geq 0$

For $z=0$, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 0$

For $0 < z < c$, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c} < 1$

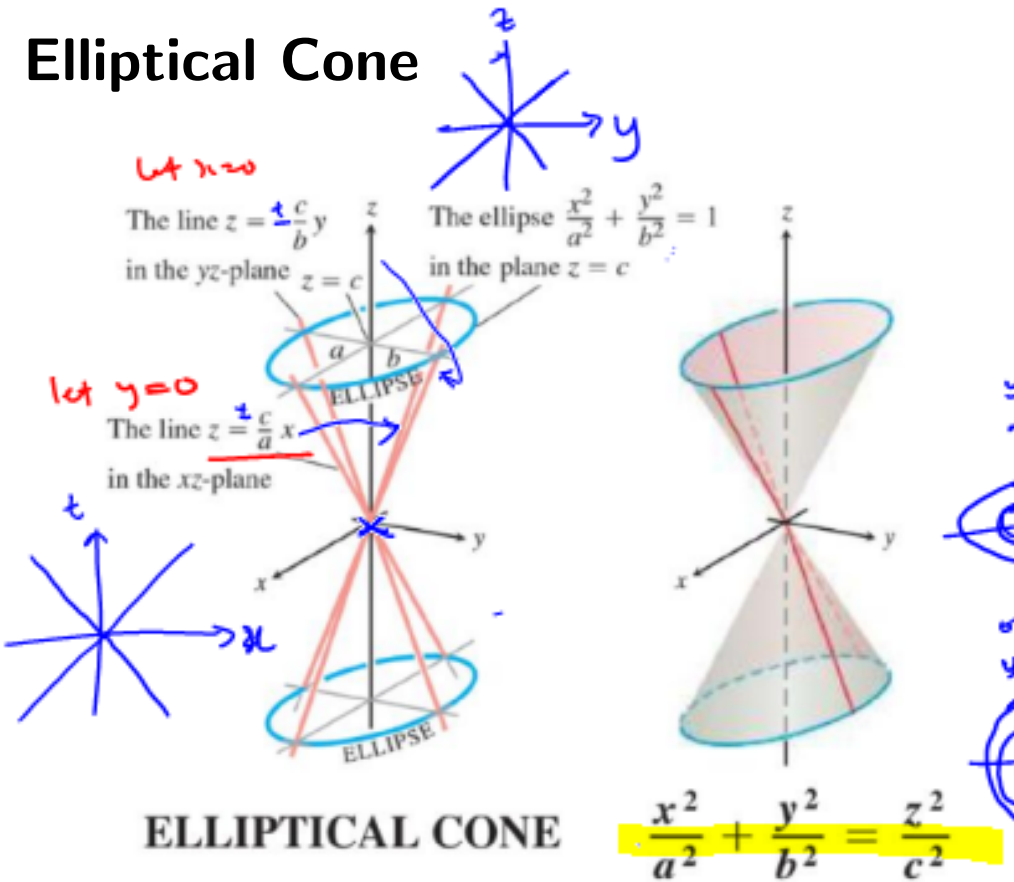
For $z = c$, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ← ellipse.

For $z > c$, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c} > 1$

$a = b = c = 1$ (Previous example in slide 4)

Quadric Surfaces

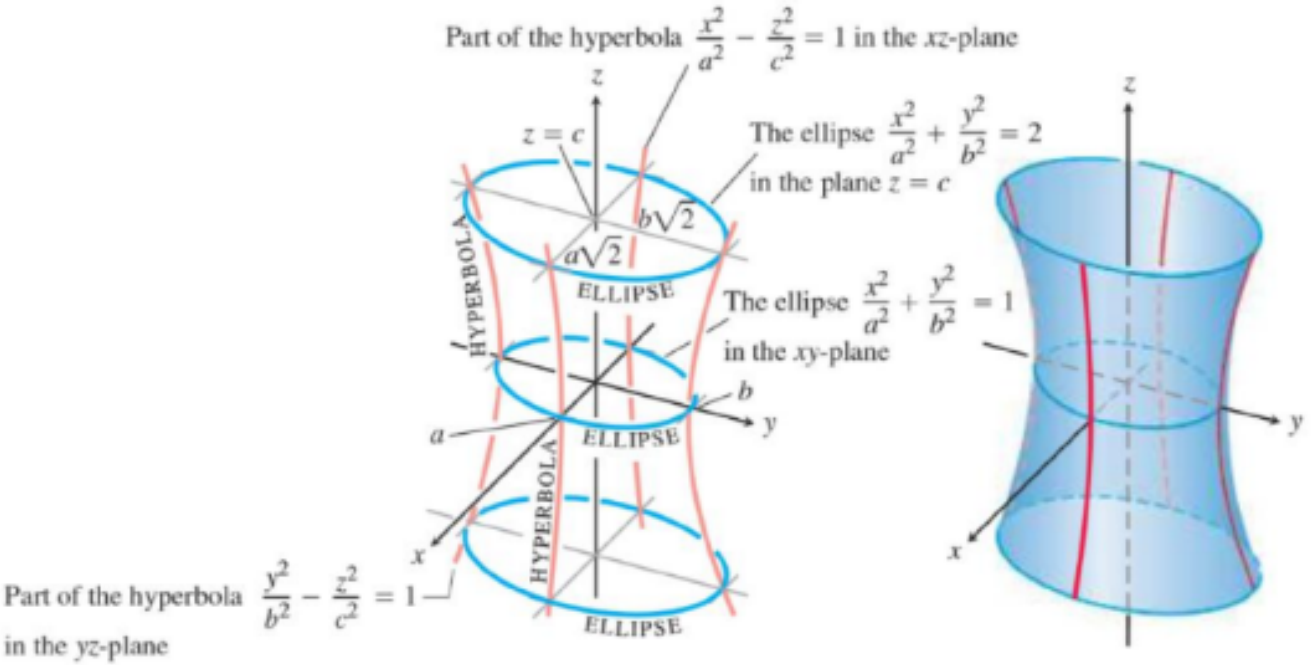
Elliptical Cone



ELLIPTICAL CONE

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$$

Hyperboloid of One Sheet

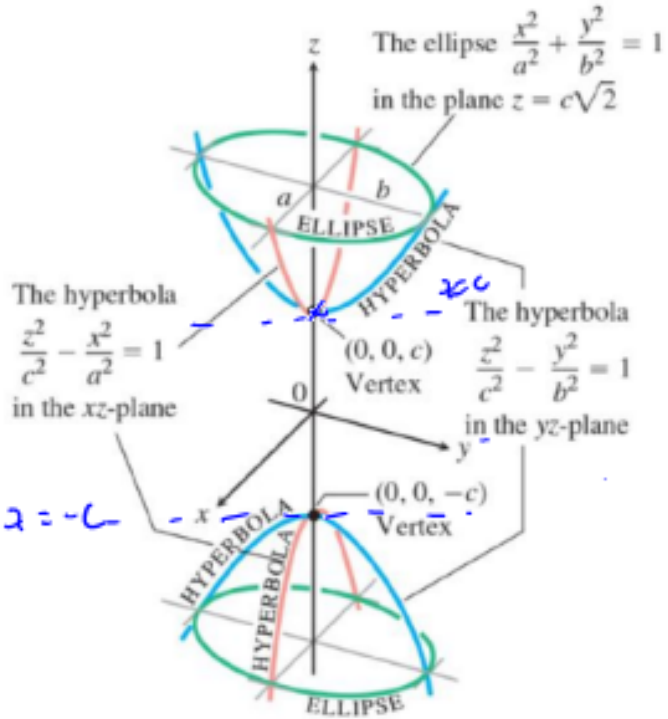


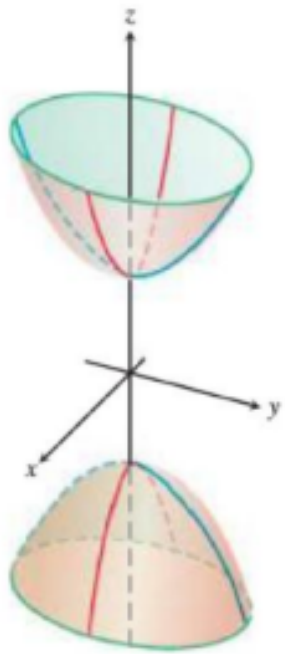
for $t=0$,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

HYPERBOLOID OF ONE SHEET $\quad \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$

Hyperboloid of Two Sheets





For $z=c$, $\frac{x^2}{a^2}+\frac{y^2}{b^2}=0$

For $z=-c$, $\frac{x^2}{a^2}+\frac{y^2}{b^2}=0$

For $z>0$, $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1>0$

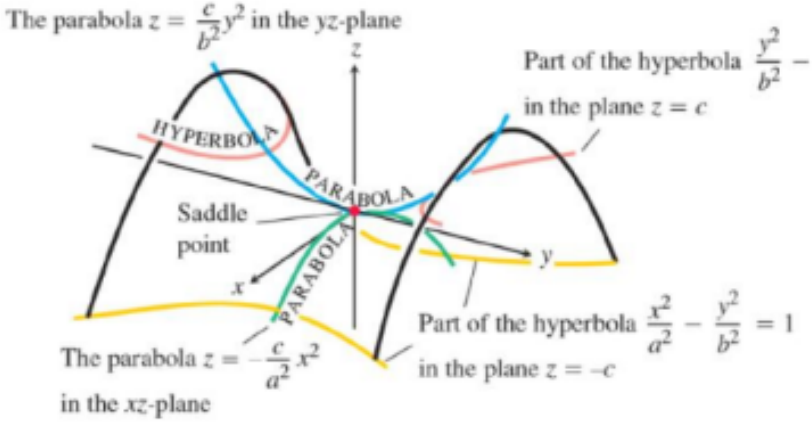
as $z\uparrow$ $\frac{z^2}{c^2}-1\uparrow$

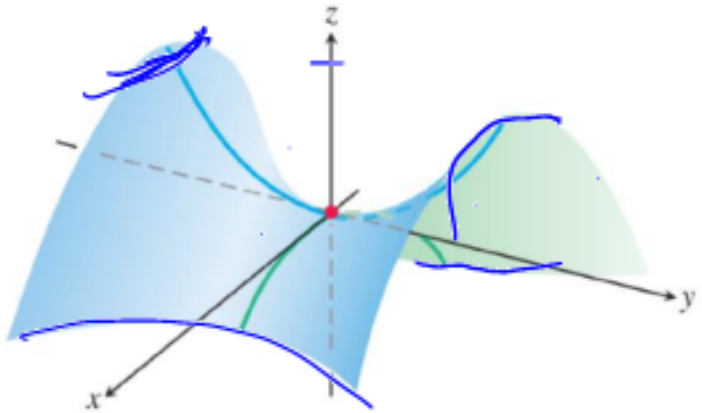
HYPERBOLOID OF TWO SHEETS $\frac{z^2}{c^2}-\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ $\Rightarrow$ $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{z^2}{c^2}-1\geq 0$

$z\leq -c,\ z\geq c$ $\Rightarrow \frac{z^2}{c^2}\geq 1$

$\Leftarrow$ $\Rightarrow x^2\geq c^2$

Hyperbolic Paraboloid





HYPERBOLIC PARABOLOID $\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}, \; c > 0$

Quadric Surfaces

Example: Identify the surface given by the equation

$$x^2 + y^2 + 4z^2 - 2x + 4y + 1 = 0.$$

$x^2 - 2x + y^2 + 4y + 4z^2 = -1$

$(x-1)^2 - 1 + (y+2)^2 - 4 + 4z^2 = -1$

$(x-1)^2 + (y+2)^2 + 4z^2 = 4$

$\frac{(x-1)^2}{4} + \frac{(y+2)^2}{4} + z^2 = 1$

[Ellipsoid centered at $(1,-2,0)$]

$z=0$, $(x-1)^2+(y+2)^2=4$. $z=1$, $(x-1)^2+(y+2)^2=0$

